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BCAC591 — PHP with MySQL | Semester V

PROJECT DETAILS REPORT

1. Project Title

Campus Resolve — Student Complaint Management System

2. Problem Statement

Students frequently encounter facility, infrastructure, and equipment-related problems across campus — such as malfunctioning classroom fans, non-functional lab computers, water cooler breakdowns, and damaged lab equipment — but lack a structured, transparent mechanism to report them and track resolution. Traditional verbal or manual complaint processes often result in untracked tickets, lack of accountability, and zero visibility for students regarding repair timelines. Campus Resolve provides a centralized, role-based platform enabling students to report grievances with location details and monitor progress through an audit timeline, while campus administrators efficiently prioritize, investigate, update, and resolve complaints with verifiable accountability.

3. User Roles

The system defines two distinct roles, fulfilling the BCAC591 project specification:

- **Student (Complainant)** — Registers with student credentials and department; signs in to access a personal workspace dashboard; submits categorized complaints specifying location, description, and optional file attachments; monitors the real-time status and historical timeline of their own complaints; and manages their account profile.
- **Administrator (Campus Operations / Authority)** — Authenticates via a separate administrative portal; oversees campus-wide operations through an analytics dashboard (reviewing pending queues, active tickets, resolution rates, and 7-day activity charts); reviews all submitted grievances across departments; updates ticket statuses (*Pending → In Progress / Under Review → Resolved / Rejected*) with official remarks; and monitors registered student accounts.

4. Module List — Mapped to This Project

Every mandatory module from the project notice, mapped specifically to this project's subject matter:

#	Mandatory Module	How It Applies to Campus Resolve	Status
1	**Authentication**	Student registration (`auth/register.php`) with 12 academic departments, separate Student Login (`auth/login.php`), separate Admin Login (`auth/admin_login.php`), and secure Sign Out (`auth/logout.php`). Passwords hashed using Bcrypt via `password_hash()` .	Implemented
2	**Role-Based Access**	Two distinct roles (`student` , `admin`) enforced via session authentication and middleware guards (`includes/auth_check_student.php` , `includes/auth_check_admin.php`). Each role has its own dedicated dashboard, navigation sidebar, and isolated permissions.	Implemented

#	Mandatory Module	How It Applies to Campus Resolve	Status
3	**Dashboard**	**Student Dashboard** (`student/dashboard.php`) shows real-time personal metrics (total, pending, in-progress, resolved, resolution rate %), time-aware greeting, and recent complaints table. **Admin Dashboard** (`admin/dashboard.php`) displays campus totals, pending action counts, registered student count, a 7-day complaint activity bar chart, and a "Needs Attention" queue.	Implemented
4	**Core Entity CRUD**	The core entity is **Complaint** (Grievance Ticket). Students can create (submit grievances) and view their own tickets. Admins can view all campus grievances, update statuses, record official administrative remarks, and track resolution lifecycles.	Partially Implemented
5	**Listing Page**	**Student Complaints List** (`student/complaints.php`) displays all tickets submitted by the logged-in student. **Admin Complaints Register** (`admin/complaints/list.php`) displays all institutional tickets with search (by complaint code, student name, or title) and filtering (by category and status).	Partially Implemented
6	**Detail View**	Dedicated screen for individual grievances (`student/complaint_details.php` and `admin/complaints/detail.php`) displaying complaint code, student metadata, category, location, full description, attachment, and chronological event history from `complaint_updates` .	Partially Implemented
7	**Status Workflow**	Every complaint moves through distinct sequential states: *Pending → In Progress / Under Review → Resolved (or Rejected)*. State transitions create audit log entries in `complaint_updates` along with administrator remarks. Only administrators can transition status.	Partially Implemented
8	**Admin Panel**	Dedicated administrative workspace featuring admin authentication, institutional KPIs, 7-day activity visual chart, high-priority ticket queue, all-complaints register, and user directory oversight.	Partially Implemented
9	**Form Validation**	Server-side validation in PHP verifies required fields, email formatting, department verification against `get_departments()` , password length and confirmation match, and CSRF token integrity via `validate_csrf()` . Client-side validation includes interactive password strength metering.	Implemented
10	**Flash Messages**	Centralized session-based flash messaging (`set_flash()` , `get_flash()`) rendered dynamically as client-side animated toast notifications (`window.showToast()`) on sign-in, account creation, sign-out, or permission checks.	Implemented
11	**Profile Management**	Student Profile (`student/profile.php`) and Admin Profile (`admin/profile.php`) interfaces to inspect account metadata (user code, institutional email, registered department) and update personal details.	Partially Implemented

Optional modules included, since they genuinely apply to this project:

- **Basic Reports / Counts & Analytics** — Implemented on Admin and Student dashboards via dynamic SQL aggregations calculating complaint volumes by status, resolution percentage rates, student user totals, and a 7-day intake activity breakdown.
- **Audit Trail / Activity Timeline** — Schema and seed implemented via `complaint_updates` table tracking historical status transitions, timestamped administrator remarks, and responsible admin IDs for complete transparency.

- **File / Evidence Upload (File Handling)** — Planned feature supported by database schema (`attachment` column in `complaints`) and server storage (`uploads/complaints/`) allowing students to upload photo evidence of damaged items or facilities.
- **Export Reports (CSV)** — Planned administrative utility (`admin/complaints/export.php`) to export campus complaint records to CSV format for administrative archiving and committee reporting.

5. Database Tables (Draft / Active Schema)

The database `campus_resolve` is configured using MySQL with foreign key constraints and `utf8mb4` encoding:

`Table: users`

Column	Type / Notes
`id`	INT, Primary Key, Auto Increment
`user_code`	VARCHAR(20), Unique *(e.g. 'ST-2016', 'AD-001')*
`name`	VARCHAR(100), Not Null
`email`	VARCHAR(120), Unique, Not Null
`password`	VARCHAR(255), Not Null — stored using `password_hash()` (Bcrypt)
`role`	ENUM('student', 'admin'), Not Null, Default 'student'
`department`	VARCHAR(100), Default NULL *(enforces 12 institutional academic departments)*
`status`	ENUM('active', 'inactive'), Not Null, Default 'active'
`created_at`	DATETIME, Default CURRENT_TIMESTAMP

`Table: complaints`

Column	Type / Notes
`id`	INT, Primary Key, Auto Increment
`complaint_code`	VARCHAR(20), Unique *(e.g. 'CR-1042')*
`user_id`	INT, Foreign Key → users.id (ON DELETE CASCADE)
`title`	VARCHAR(200), Not Null
`category`	ENUM('Infrastructure', 'IT', 'Facilities', 'Laboratory', 'Other'), Not Null
`location`	VARCHAR(150), Not Null *(e.g. 'Block B · Room 204')*
`description`	TEXT, Not Null
`attachment`	VARCHAR(255), Nullable *(relative path to uploaded evidence file)*
`status`	ENUM('Pending', 'In Progress', 'Resolved', 'Rejected'), Default 'Pending'
`created_at`	DATETIME, Default CURRENT_TIMESTAMP
`updated_at`	DATETIME, Default CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

`Table: complaint_updates`

Column	Type / Notes
`id`	INT, Primary Key, Auto Increment
`complaint_id`	INT, Foreign Key → complaints.id (ON DELETE CASCADE)
`admin_id`	INT, Nullable, Foreign Key → users.id (ON DELETE SET NULL)
`status`	VARCHAR(50), Not Null *(e.g. 'Submitted', 'Under Review', 'In Progress', 'Resolved')*
`remark`	TEXT, Not Null

Column	Type / Notes
`created_at`	DATETIME, Default CURRENT_TIMESTAMP

6. Page List

#	Page / Screen	Accessible By	Purpose	Status
1	**Public Landing Page** `index.php`	Public	Introduces the system, explains 3-step workflow, displays live tracking demo, and directs users to role sign-in.	Implemented
2	**Student Sign In** `auth/login.php`	Public	Student credential authentication with CSRF protection, password toggle, and session creation.	Implemented
3	**Admin Sign In** `auth/admin_login.php`	Public	Dedicated administrative authentication portal restricted strictly to accounts with `role = 'admin'`.	Implemented
4	**Student Registration** `auth/register.php`	Public	New student onboarding with academic department dropdown selection (12 departments), Bcrypt hashing, and auto-generated user code.	Implemented
5	**Sign Out Handler** `auth/logout.php`	Authenticated	Safely destroys session, unsets session cookies, and redirects to login with flash confirmation.	Implemented
6	**Student Dashboard** `student/dashboard.php`	Student	Student workspace displaying personal KPI counters (total, pending, in-progress, resolved), resolution rate %, and recent complaints table.	Implemented
7	**Submit Complaint** `student/submit_complaint.php`	Student	Form to lodge a new complaint specifying title, category dropdown, location, description, and file attachment.	Planned
8	**My Complaints Listing** `student/complaints.php`	Student	Filterable and searchable table of all grievances submitted by the logged-in student.	Planned
9	**Complaint Detail View** `student/complaint_details.php`	Student	Comprehensive single complaint view showing issue breakdown, location, evidence, and chronological status timeline.	Planned
10	**Student Profile** `student/profile.php`	Student	Profile management page allowing students to view credentials and update their name and department.	Planned
11	**Admin Dashboard** `admin/dashboard.php`	Admin	Centralized administrative analytics hub with campus-wide statistics, resolution percentage, 7-day intake chart, and "Needs Attention" queue.	Implemented
12	**All Complaints Register** `admin/complaints/list.php`	Admin	Institutional complaints management table with search, category filtering, and status filtering.	Planned

#	Page / Screen	Accessible By	Purpose	Status
13	**Admin Complaint Review** `admin/complaints/detail.php`	Admin	Detailed ticket inspection screen with controls to transition status and enter official administrative remarks.	Planned
14	**Update Status Action** `admin/complaints/update_status.php`	Admin	POST handler updating complaint status and creating an audit record in `complaint_updates`.	Planned
15	**User Directory** `admin/users.php`	Admin	Overview of registered students across departments with account status and ticket counts.	Planned
16	**Export Complaints (CSV)** `admin/complaints/export.php`	Admin	Streams server-generated CSV file containing grievance records for official reporting.	Planned
17	**Admin Profile** `admin/profile.php`	Admin	Administrative profile and security configuration screen.	Planned

7. File Structure

```
campus-resolve/
│
├── config/
│   └── db_connect.php              # PDO MySQL connection with error handling & auto-setup
│
├── includes/
│   ├── auth.php                    # Session handling, role helpers, and CSRF token utilities
│   ├── auth_check_student.php      # Access guard: enforces authenticated student role
│   ├── auth_check_admin.php        # Access guard: enforces authenticated administrator role
│   ├── helpers.php                 # Utilities: XSS escaping (e()), status badges, get_departments()
│   ├── header.php                  # Shared top app navigation bar with user avatar & breadcrumb
│   ├── footer.php                  # Shared app footer and dynamic toast notification trigger
│   ├── student-sidebar.php         # Navigation sidebar for student workspace
│   └── admin-sidebar.php           # Navigation sidebar for administrative workspace
│
├── assets/
│   ├── css/
│   │   └── styles.css              # Application design system: tokens, typography, layouts, badges
│   └── js/
│       └── app.js                  # Client utilities: mobile nav, toast alerts, password toggle
│
├── icons.svg                       # Shared SVG icon symbol sprite (check, bell, chevron, etc.)
│
├── auth/
│   ├── register.php                # Student account registration with 12-department dropdown
│   ├── login.php                   # Student sign-in portal with session initialization
│   ├── admin_login.php             # Dedicated administrative sign-in portal
│   └── logout.php                  # Secure session destruction and redirect
│
├── student/
│   ├── dashboard.php               # Student home: personal metrics, recent tickets, quick links
│   ├── submit_complaint.php        # [Planned] Grievance submission form with evidence upload
│   ├── complaints.php              # [Planned] Student's personal complaints listing & filter
│   ├── complaint_details.php       # [Planned] Detail view with chronological update timeline
│   └── profile.php                 # [Planned] Student profile and department update
│
├── admin/
│   ├── dashboard.php               # Operations dashboard: campus KPIs, 7-day chart, active queue
│   ├── complaints/
│   │   ├── list.php                # [Planned] All campus complaints with search & filters
│   │   ├── detail.php              # [Planned] Admin complaint inspection & remark entry
│   │   ├── update_status.php       # [Planned] POST handler for status transitions & updates
│   │   └── export.php              # [Planned] CSV data export of campus complaints
│   ├── users.php                   # [Planned] Student user directory across departments
│   └── profile.php                 # [Planned] Admin profile management
│
├── database/
│   ├── schema.sql                  # DDL: users, complaints, complaint_updates tables
│   └── seed.sql                    # Seed dataset: initial admin, sample students & complaints
│
├── uploads/
│   └── complaints/                 # [Planned] Storage folder for uploaded complaint evidence
│
└── index.php                       # Public landing page with system walkthrough & role entry
```